

Enhancing E-Commerce Performance Through PDP Optimization

Insights from NOVICA's Amazon Listings

By Team OptiMystics

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Executive Summary

Our project focuses on how NOVICA's product detail pages (PDPs) on Amazon influence sales performance across its top and bottom-selling SKUs. While the Exploratory Data Analysis revealed clear performance gaps, especially in A/A+ content adoption, multimedia use, and rating quality, these trends raised an important question: Which content attributes actually drive conversions, and how can NOVICA use them to improve underperforming listings?

To understand further, our team conducted a structured analysis focused on five key areas i.e., the impact of A/A+ content, the role of videos and images, the relationship between customer ratings and sales, category-specific differences in content effectiveness, and the identification of "hidden gem" SKUs with strong attributes but weak performance. Using descriptive comparisons, statistical tests, and regression modelling, we quantified the influence of each PDP element and evaluated how these effects vary across product categories such as Jewellery, Home Decor, and Sculpture.

On a high level, our analysis showcased that several content features, especially A+ content, higher ratings, and richer multimedia, are consistently associated with stronger sales performance. However, the strength of these relationships varies by category, suggesting that a single content strategy is not optimal for all product types. We also identified a subset of low-selling SKUs whose content profiles closely resemble top performers, indicating untapped potential for targeted improvements.

To ensure this report reads as a fully standalone document, the analysis summarized here is presented independently of earlier exploratory steps. The goal is to clearly highlight the core patterns in NOVICA's Amazon catalog and explain which PDP elements most consistently influence conversion performance across product categories.

Overall, the patterns across A/A+ content, media richness, ratings, and keyword strength demonstrate how strongly PDP elements shape shopper decisions on Amazon. These insights point toward practical actions NOVICA can take to strengthen visibility and improve conversions without altering its core brand identity. By focusing on elements that consistently relate to stronger sales and identifying SKUs that already exhibit high-quality content but low performance, NOVICA can prioritize updates that offer the quickest and most measurable lift.

This report includes the recommendations of NOVICA's marketing and merchandising teams as per our findings through the analysis.

Data and Methodology

Dataset

- 2,500 top-selling and 2,500 low-selling Amazon SKUs
- For each SKU we used:
 - **Sales:** total units sold
 - **Content:** A/A+ content status, total images, total videos
 - **Customer signals:** average rating, review count

- **Product attributes:** category / classification
- **Text features:** extracted keywords from titles, bullets and descriptions to support keyword analysis.

Methodology for analysis

- Exploratory data analysis and visualizations
- Statistical tests: t-tests, Mann–Whitney U test, ANOVA, Pearson's correlation test
- **OLS regression** with heteroskedasticity-robust (HC3) standard errors
- Simple **classification/screening** logic to identify hidden gems
- Custom **keyword impact scoring**, combining frequency, correlation with sales, sentiment, and machine learning derived importance weights using a Random Forest regression model

Diagnostics showed heteroskedastic residuals and some multicollinearity in interaction models, so coefficients are interpreted **directionally and in relative terms**, not as precise forecasts.

Analysis and Findings

3.1 Objective 1 – Impact of A/A+ Content and Media

Question: How do A/A+ content, images, and videos relate to unit sales?

Model Specification:

- **Dependent variable:** capped total quantity (to limit outlier influence)

- **Predictors:** A/A+ content indicators, number of images, number of videos, rating, review count, category dummies
- **Method:** OLS with HC3 robust standard errors

Key Results:

- **Model fit:** $R^2 \approx 0.37$ (Adj. $R^2 \approx 0.36$)
- **A+ content:** SKUs with A+ content show ~85% higher sales relative to the baseline.
- **No enhanced content:** associated with ~16% lower sales.
- **Media richness:** additional images and videos both have positive, significant effects.
- **Ratings:** minimal effect once content and category controls are included.

Takeaway:

A+ content and richer media are strong predictors of higher sales. Ensuring each PDP includes enhanced content, sufficient images, and at least one video appears critical for improving performance. Refer to Exhibit 1 for a visual comparison of average sales performance across SKUs with and without enhanced A/A+ content.

3.2 Objective 2 – Keyword Impact on Novica Product Sales

Question: Which keywords in titles and descriptions are most associated with higher sales, and how can Novica standardize them?

Approach:

Extracted descriptive and brand-relevant keywords from titles and metadata, then

calculated each term's frequency, correlation with sales, and concentration in top sellers. These were combined into a normalized composite "impact score" and grouped into High, Medium, and Low Priority clusters.

High-Impact Keywords:

The strongest performers emphasize craftsmanship and materials: *handmade*, *artisan*, *silver*, *sterling*, and related terms such as *artisans*, *talented*, *artists*, *designers*, *birthstone*. These show the highest correlations with sales and consistently appear in top-selling SKUs. See Exhibit 6 for a detailed keyword impact table illustrating how high-impact keywords cluster in top-selling listings.

Medium-Impact Keywords:

Product-type terms (*ring*, *earrings*, *quality*, *floral*) show positive but more context-dependent effects. They work best when paired with the high-impact brand and material terms.

Low-Impact Keywords:

Generic descriptors (*painted*, *diam*, *guatemala*, *card*, *piece*) have low or negative impact scores. They add description but do not meaningfully support conversion and can dilute clarity when overused.

Takeaways:

- Lead titles and descriptions with high-impact terms tied to Novica's brand: *handmade*, *artisan*, plus accurate material words such as *silver* or *sterling*.

- Reinforce artistry in titles and description using supporting terms like *artisans*, *artists*, and *designers*. See Exhibit 6 for a detailed keyword impact table illustrating how high-impact keywords cluster in top-selling listings.
- Limit low-impact filler and use geographic mentions only when they strengthen the artisan story.
- Create category-specific keyword templates combining:
 - High-impact brand terms
 - Material descriptors
 - Product type
 - Occasion or use case
- Adopt shorter, high-impact titles on Amazon to improve clarity, SEO performance, and conversion.

This keyword analysis turns Novica’s brand positioning into a **data-backed copywriting guideline**.

3.3 Objective 3 – Number of Videos and Sales Performance

Question: Does adding more videos materially affect sales?

Methods

- Compared average video counts between top- and bottom-selling SKUs.
- Estimated a regression with log(quantity) as the dependent variable and video count, A+ indicator, rating, and a video × A+ interaction.

Key Results

- **Video usage:**
 - Top sellers average 1.29 videos; bottom sellers average 0.94.
 - This reflects a 37% higher video count for top sellers.
 - Difference is statistically significant ($t \approx 11.26$, $p \approx 2.9 \times 10^{-28}$).
- **Regression:**
 - Video count is positive and significant.
 - The video $\times$ A+ interaction is unstable due to multicollinearity and provides no additional explanatory value.
 - Model $R^2 \approx 0.05$, consistent with a single-attribute focus.

Takeaway

Videos are a meaningful differentiator. While not a standalone driver, SKUs without video consistently underperform, especially in categories where product usage or movement matters. Refer to Exhibit 2 for a visual comparison of video usage across top- and bottom-selling SKUs, illustrating the clear differences in how frequently videos are included and reinforcing the conclusion that video presence serves as a meaningful differentiator in sales performance.

3.4 Objective 4 – Ratings and Reviews

Question: Do small increases in already-high ratings affect sales, and how does review volume relate to performance?

Methods

- Regressions linking sales to rating bands and total reviews, with category controls.

- Boxplots comparing sales across rating buckets.

Key Results

- Ratings are tightly clustered between 4 and 5 stars, limiting explanatory value.
- Within this narrow band, rating differences do not significantly affect sales once other factors are controlled.
- Review count shows a positive relationship with sales, though partly mechanical because high-selling SKUs accumulate reviews.

Takeaway

High ratings are necessary but offer limited marginal impact in Novica's catalog. Moving a product from 4.6 to 4.8 stars is far less influential than improving PDP content or media. Review generation remains helpful but should be treated as a secondary lever. Refer to Exhibit 3 for a visual comparison of rating distributions across top- and bottom-selling SKUs, reinforcing the conclusion that limited variance in ratings reduces their explanatory power in the model.

3.5 Objective 5 – Category-Level Content Impact

Question: Are content levers equally effective across categories?

Methods

- MANOVA with content features (A+/A presence, images, videos, ratings) as outcomes and category plus sales group as factors.

- Category-augmented OLS model with total quantity as the dependent variable and category dummies, A+ content, images, videos, and ratings as predictors.

Key Results

- MANOVA tests (Wilks, Pillai, Hotelling–Lawley) are all highly significant, showing that category shapes how content relates to performance.
- In the OLS model ($R^2 \approx 0.084$; Adj. $R^2 \approx 0.059$):
 - A+ content adds ~22 units of sales on average.
 - Each additional image contributes ~2 units.
 - Each additional video contributes ~3–4 units.
- Category effects vary substantially, with some categories selling more regardless of similar content levels.

Takeaway

Content levers do not perform uniformly. Categories require tailored guidance because the ROI of images, videos, and A+ content differs across Jewelry, Tableware, Sculpture, and other segments. Refer to Exhibit 4 for a visual summary of how the influence of A+ content, images, videos, and ratings differs across categories, highlighting why category-specific optimization strategies are necessary.

3.6 Objective 6 – Identifying “Hidden Gem” SKUs

Question:

Which low-selling SKUs have “top seller-like” PDPs and may simply need more visibility?

Approach:

- Labeled SKUs by sales quartile (top vs. bottom).
- Compared A+ use, images, videos, and ratings between quartiles.
- Built a simple **screening rule** that flags low-selling SKUs whose content metrics meet or exceed median values for top sellers.
- Examined basic correlations among sales, A+ count, rating, images, and videos.

Key results:

- Pearson correlations with sales:
 - Videos: **strongest positive correlation** among the content variables
 - Images: modest positive correlation
 - A+ count and rating: very small correlations (consistent with earlier models)
- The screening logic identifies **≈750–753 SKUs** as “**hidden gems**”:
 - Low observed sales
 - Strong ratings
 - Reasonable or high numbers of images and sometimes videos
 - Often in visually rich categories like **glassware, vases, and decorative bowls**
- Many of these SKUs already embody Novica’s brand story and craftsmanship; they are simply under-discovered.

Takeaway:

These hidden gems form a **ready-made campaign backlog** for Novica’s performance

marketing and merchandising teams. Because content is already strong, lifting visibility (ads, placements, bundles) is likely to have **above-average incremental impact**. Refer to Exhibit 5 for a correlation heatmap that visualizes the relationships among sales and key PDP attributes, highlighting the relative strength of videos and images compared to the weaker associations observed for ratings and A+ content.

Recommendations for Novica

As per our analysis, following are the recommendations we would provide to NOVICA:

1. Expand A+ content coverage

Prioritize SKUs with strong ratings and steady traffic but no A+ content, focusing first on higher-margin categories.

2. Establish minimum media standards by category

Set clear baselines (for example, ≥ 8 images and ≥ 1 video for hero SKUs in Tableware, Décor, and Jewellery) to ensure consistency and conversion readiness.

3. Promote hidden-gem SKUs

Activate marketing (ads, email, onsite placements) for the ~750 SKUs with strong content but low sales, and measure performance uplift against a control group.

4. Use category-specific content playbooks

- *Jewellery*: emphasize material, craftsmanship, and strong keywords.
- *Tableware/Glassware*: highlight use, scale, and motion through video.
- *Décor/Sculpture*: focus on artisan story, dimensions, and scale clarity.

5. Standardize keyword and copy templates

Build reusable keyword frameworks tailored by category and update them using periodic keyword analysis to match shifts in catalog performance and Amazon search trends.

6. Treat ratings as hygiene, not a growth lever

Preserve current high ratings through service quality, but direct optimization efforts toward content, media, and keywords where the data shows greater sales impact.

Limitations

Sales performance in this dataset is shaped not only by PDP content but also by external factors such as traffic, advertising exposure, and click-through behavior. Because impression-level data and ad spend information were unavailable, our analysis cannot fully distinguish conversion effects from visibility effects. In addition, the study does not incorporate historical pricing or promotional activity, which may explain part of the variation observed between top- and bottom-selling SKUs. Finally, the keyword analysis reflects current catalog usage, meaning it may under-represent emerging or experimental keyword strategies that have not yet been adopted widely.

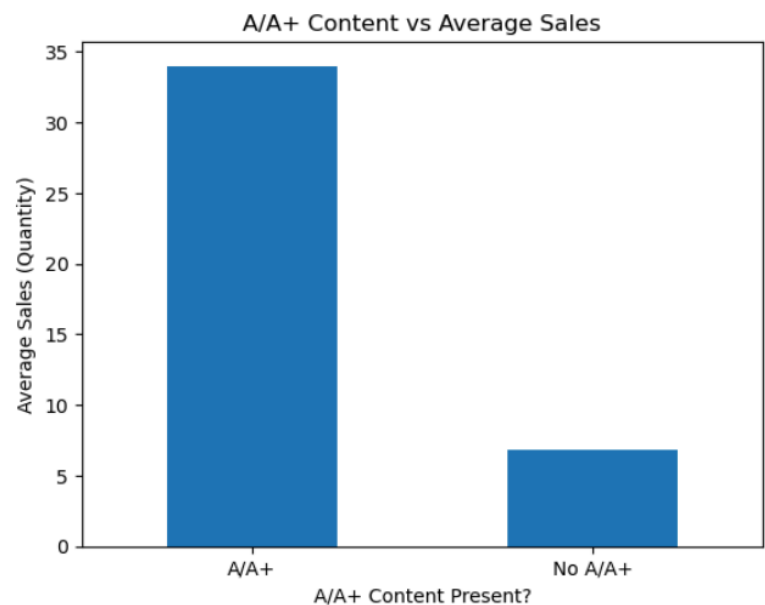
Further Analysis Scope

Future analysis should integrate traffic and advertising data to clearly separate visibility-driven drivers from true conversion levers. Controlled experiments—such as adding A+ content or videos to matched test and control SKU pairs—would provide

causal evidence of content impact. Additionally, extending the keyword scoring framework with search-term reports and sponsored ads data would offer a more complete view of how on-page copy aligns with customer search behavior, enabling NOVICA to optimize PDP content and paid visibility strategies more cohesively.

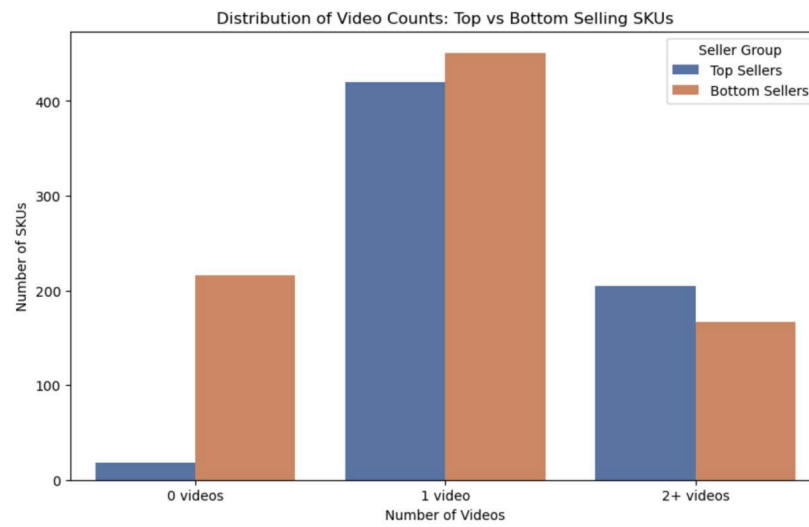
References and Exhibits

Exhibit 1



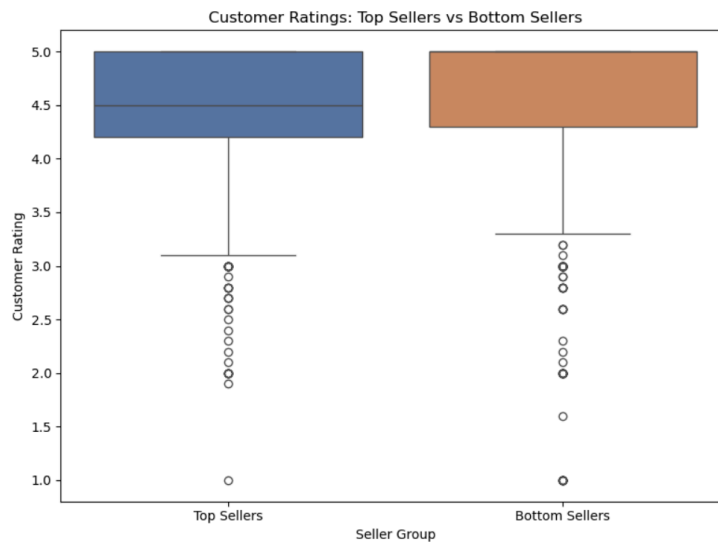
** Exhibit 1 visualizes the difference in average sales between SKUs with and without enhanced A/A+ content. The chart clearly shows that SKUs featuring A+ content outperform those without it by a substantial margin, reinforcing the model’s finding that enhanced content is one of the strongest predictors of higher sales. The visual also highlights how richer media—such as additional images and videos—correlates with better performance, illustrating the importance of a complete and compelling product detail page in driving conversions.*

Exhibit 2



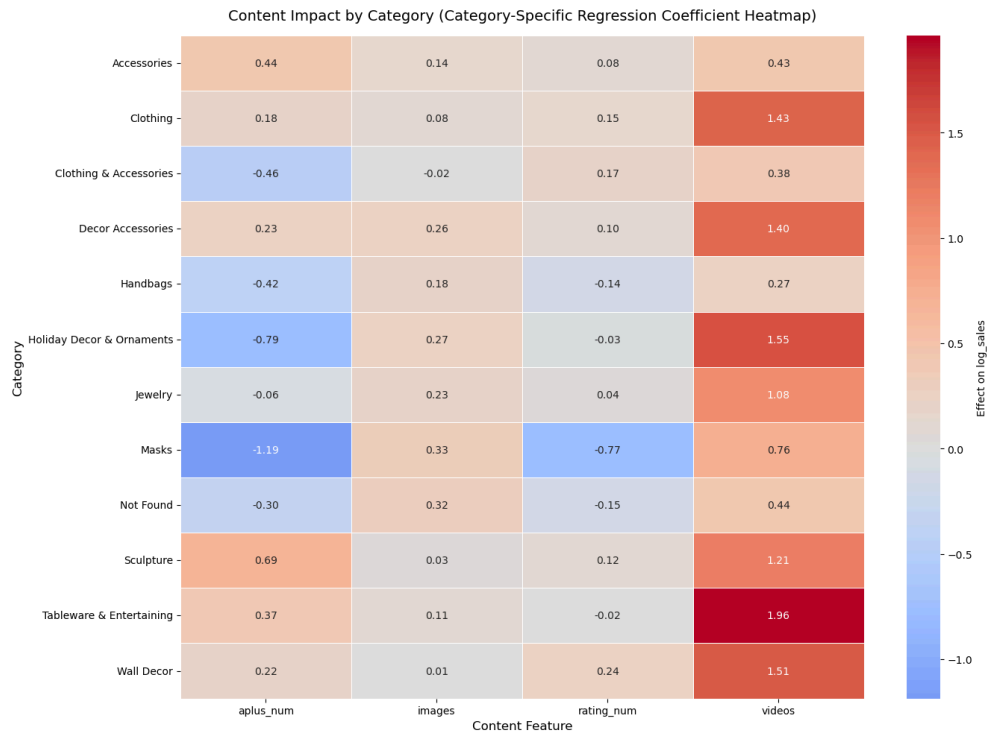
* Exhibit 2 compares video usage between top- and bottom-selling SKUs, showing a noticeable gap in both the incidence and quantity of videos. Top sellers include significantly more videos on average, underscoring the role of video in demonstrating product scale, movement, and use context. This visual supports the regression findings by illustrating that although videos are not the sole determinant of success, they serve as a consistent differentiator, especially in categories where physical interaction or visual demonstration influences purchasing decisions.

Exhibit 3



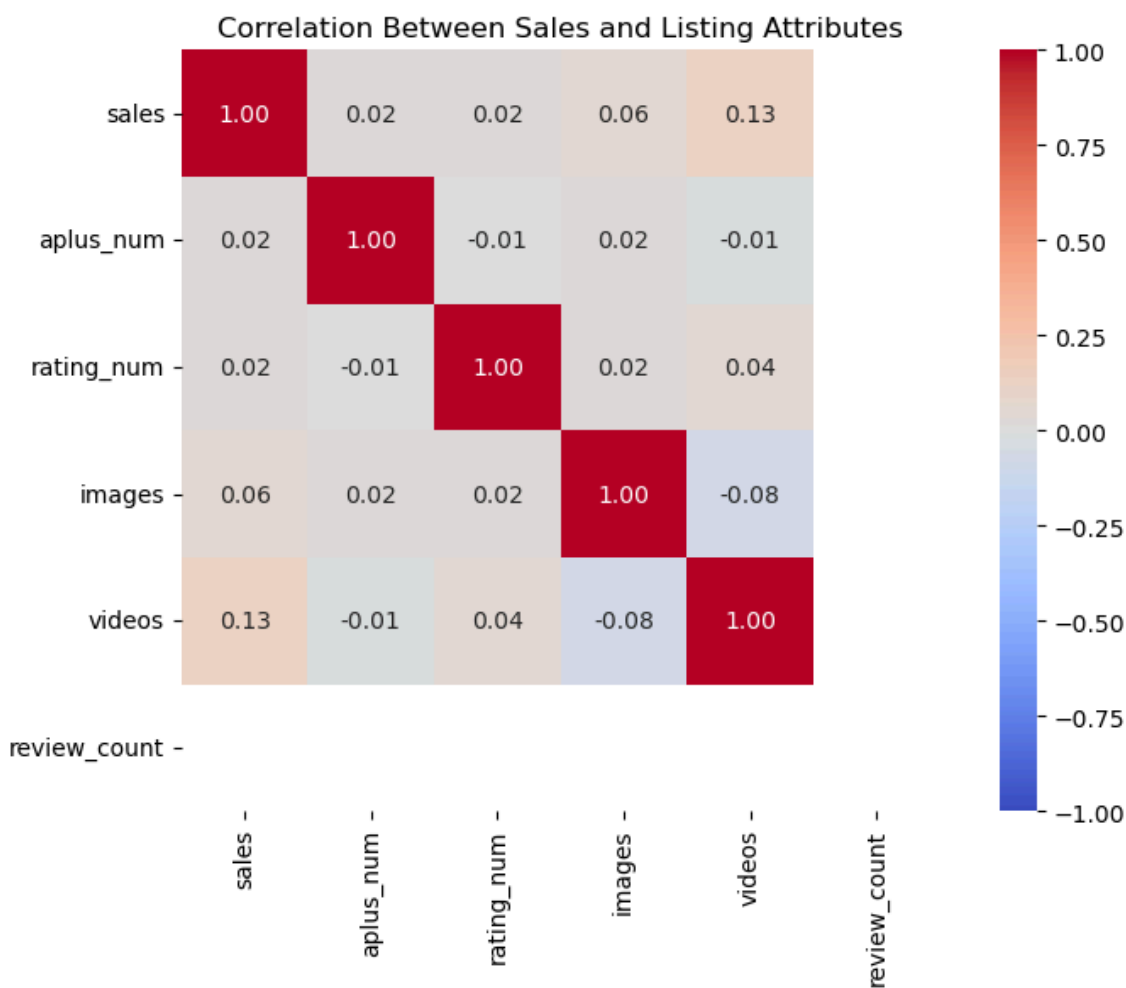
* Exhibit 3 displays rating distributions across sales groups, revealing that most SKUs—top and bottom sellers alike—cluster tightly between 4 and 5 stars. This limited variance visually confirms why ratings contribute little explanatory power in the regression models despite being generally high. The chart also shows that while review count increases with sales volume, the rating scores themselves do not distinguish high performers from low performers, supporting the conclusion that ratings function more as a hygiene factor than a primary growth lever.

Exhibit 4



* This heatmap illustrates how the effectiveness of PDP content varies considerably across categories. Warmer colors represent stronger positive associations with sales, while cooler colors reflect weaker or even negative effects. For example, videos appear especially influential in categories such as Tableware & Entertaining and Holiday Décor, whereas images and A+ content contribute more meaningfully in Jewelry and Accessories. These patterns reinforce the finding that a single, uniform content strategy is insufficient for NOVICA's catalog; instead, each category benefits from tailored optimization priorities based on how customers evaluate and compare products within that segment.

Exhibit 5



* Exhibit 5 presents a correlation heatmap that highlights how different PDP attributes relate to sales. The visual shows that videos have the strongest positive correlation with sales among the content variables, followed by images, while ratings and A+ content display weaker relationships. These correlation patterns help explain why many low-selling SKUs with strong media profiles were flagged as “hidden gems.” The heatmap thus supports the idea that improving visibility—not content quality—is likely the highest-impact lever for these SKUs

Exhibit 6

	SKU	Classification	keyword	impact_score	freq_scaled	corr_scaled	imp_scaled	sent_scaled	gap_vs_benchmark	priority
0	219746a77275	Jewelry	balinese	0.069684	0.058039	0.219819	0.000877	0.0	-0.065263	Low Priority
2	219746a77275	Jewelry	artisan	0.499794	0.988392	0.994476	0.016309	0.0	0.364848	High Priority
3	219746a77275	Jewelry	handmade	0.503462	1.000000	1.000000	0.013847	0.0	0.368515	High Priority
4	219746a77275	Jewelry	amethyst	0.085788	0.081947	0.260547	0.000659	0.0	-0.049158	Low Priority
5	219746a77275	Jewelry	garnet	0.072743	0.063410	0.226723	0.000840	0.0	-0.062204	Low Priority
6	219746a77275	Jewelry	cocktail	0.129563	0.156272	0.361033	0.000949	0.0	-0.005383	Medium Priority
7	219746a77275	Jewelry	ring	0.192876	0.280319	0.489962	0.001221	0.0	0.057929	Medium Priority
8	219746a77275	Jewelry	large	0.018211	0.005891	0.066883	0.000072	0.0	-0.116735	Low Priority
9	219746a77275	Jewelry	silver	0.365368	0.668919	0.786258	0.006296	0.0	0.230421	High Priority
10	219746a77275	Jewelry	sterling	0.358318	0.652807	0.775617	0.004849	0.0	0.223371	High Priority
11	219746a77275	Jewelry	purple	0.077877	0.070686	0.240403	0.000417	0.0	-0.057070	Low Priority
12	219746a77275	Jewelry	red	0.083231	0.078482	0.253573	0.000869	0.0	-0.051716	Low Priority
14	219746a77275	Jewelry	birthstone	0.213838	0.321552	0.530284	0.003518	0.0	0.078892	Medium Priority
15	219746a77275	Jewelry	traditional	0.106201	0.103604	0.314618	0.006583	0.0	-0.028745	Low Priority
16	219746a77275	Jewelry	royal	0.032218	0.015419	0.112171	0.001282	0.0	-0.102728	Low Priority
17	257441a111193	Jewelry	ring	0.192876	0.280319	0.489962	0.001221	0.0	0.057929	Medium Priority
19	257441a111193	Jewelry	handmade	0.503462	1.000000	1.000000	0.013847	0.0	0.368515	High Priority
20	257441a111193	Jewelry	sterling	0.358318	0.652807	0.775617	0.004849	0.0	0.223371	High Priority
21	257441a111193	Jewelry	silver	0.365368	0.668919	0.786258	0.006296	0.0	0.230421	High Priority
22	257441a111193	Jewelry	cocktail	0.129563	0.156272	0.361033	0.000949	0.0	-0.005383	Medium Priority

* Exhibit 6 provides a detailed breakdown of keyword impact scores, showing how high-impact terms such as “handmade,” “artisan,” “silver,” and “sterling” appear far more frequently in top-selling SKUs and carry stronger positive associations with sales. In contrast, low-impact generic descriptors appear inconsistently and show minimal correlation with performance. This visual reinforces the text-based analysis by translating keyword effectiveness into a quantitative display, helping identify which terms should anchor NOVICA’s copywriting templates across categories.